



DEPLOYMENT GUIDE

TravelPort

Docker · GitHub Actions CI/CD · Server Provisioning

VERSION

v1.0.0

PLATFORM

Docker + Linux VPS

CI/CD

GitHub Actions

REGISTRY

ghcr.io

Table of Contents

Deployment Guide — TravelPort v1.0.0

1	Prerequisites	3
1.1	Local Dev	3
1.2	Production Server	3
2	Docker Setup	5
2.1	docker-compose Services	5
2.2	Environment Variables	5
2.3	Build & Run	5
3	CI/CD Pipeline	7
3.1	GitHub Actions Workflow	7
3.2	Required Secrets	7
3.3	Deploy Job	7
4	Manual Deployment	9
4.1	Server Bootstrap	9
4.2	Rolling Update	9
5	Database Management	11
5.1	Auto Migrations	11
5.2	Backup & Restore	11
6	Troubleshooting	13

1

Prerequisites

Required tools for local development and production

Local Development

- **Node.js 20+** — Frontend build
- **.NET 8 SDK** — Backend build & EF tools
- **SQL Server Express** —
`localhost\SQLEXPRESS`
- **Docker Desktop** (optional) — for container testing
- **Git** — Source control

Production Server

- **Ubuntu 22.04+** (or any Linux)
- **Docker Engine 24+**
- **Docker Compose v2**
- **SSH access** — For GHA deployment
- **Ports open:** 80 (HTTP), 443 (HTTPS)

2

Docker Setup

Three-container stack: SQL Server + .NET API + Nginx/React

2.1 docker-compose Services

SERVICE	IMAGE	PORT	HEALTH CHECK
sqlserver	mcr.microsoft.com/mssql/server:2022-latest	1433 (internal)	sqlcmd SELECT 1
api	ghcr.io/{owner}/travelport-api	8080 (internal)	GET /health → 200
web	ghcr.io/{owner}/travelport-web	80 → host	HTTP 200 on /

The `web` (Nginx) container serves the React SPA and reverse-proxies `/api/*` requests to the `api` container on port 8080.

2.2 Environment Variables

Copy `.env.example` to `.env` and fill in the required values:

```
# Required
DB_SA_PASSWORD=YourStr0ngP@ssword!
JWT_SECRET=minimum-32-character-random-secret-key
JWT_ISSUER=TravelPort
JWT_AUDIENCE=TravelPortApp
ASPNETCORE_ENVIRONMENT=Production

# Email (Office365)
EMAIL_ENABLED=true
EMAIL_HOST=smtp.office365.com
EMAIL_PORT=587
EMAIL_USERNAME=noreply@yourdomain.com
EMAIL_PASSWORD=your-smtp-password

# Payment (Razorpay - optional)
RAZORPAY_ENABLED=false
RAZORPAY_KEY_ID=rzp_test_xxx
RAZORPAY_KEY_SECRET=your-secret
```

2.3 Build & Run

```
# First run – build images and start all services
cp .env.example .env
# Edit .env with your values
docker compose build
docker compose up -d

# Verify services
docker compose ps
docker compose logs api --tail=50

# App URLs
# Frontend: http://localhost
# API Swagger: http://localhost/api/swagger
```

i First Start Timing

SQL Server takes ~45 seconds to initialize. The API container waits via a healthcheck dependency. EF Core migrations and data seeding run automatically on API startup — no manual steps required.

3

CI/CD Pipeline

GitHub Actions 3-job workflow

3.1 Workflow Jobs

1

build-and-verify

Triggered on push to Development or PR. Runs dotnet build + dotnet test (backend) and npm run build (frontend). Both must pass before proceeding.

2

docker-build-push

Runs only on push to Development (not on PRs). Builds multi-stage Docker images for API and Web. Pushes tagged images to ghcr.io.

3

deploy

Requires manual approval via GitHub Environment protection rules. SSHes into production server, pulls latest images and runs docker compose up -d.

3.2 Required GitHub Secrets

SECRET NAME	VALUE
DEPLOY_HOST	Production server IP or hostname
DEPLOY_USER	SSH username (e.g., ubuntu)
DEPLOY_KEY	SSH private key (PEM format, no passphrase)
DB_SA_PASSWORD	SQL Server SA password (strong, 16+ chars)
JWT_SECRET	Minimum 32-character random string
EMAIL_USERNAME	SMTP sender email address
EMAIL_PASSWORD	SMTP password or app password
RAZORPAY_KEY_ID	Razorpay Key ID (optional)
RAZORPAY_KEY_SECRET	Razorpay Key Secret (optional)

Security Warning

Never commit secrets to appsettings.json or any tracked file. Secrets flow via GitHub Secrets → server .env file at deploy time. The .env file is in .gitignore.

3.3 Rollback

```
# SSH into server and roll back to previous image tag
docker compose pull travelport-api:previous-tag
docker compose up -d

# Or use git tag to redeploy a previous release
git checkout v1.0.0
git push origin Development # triggers pipeline
```

4

Manual Deployment

First-time server bootstrap

4.1 Server Bootstrap

```
# 1. Install Docker on Ubuntu
curl -fsSL https://get.docker.com | sh
sudo usermod -aG docker $USER

# 2. Clone repository
git clone https://github.com/YourOrg/Goibibo-AI-Assignment.git
cd Goibibo-AI-Assignment

# 3. Create .env from secrets
cat > .env <# ... all other vars
EOF

# 4. Pull pre-built images and start
docker compose pull
docker compose up -d
```

5

Database Management

5.1 Reset Database (Dev Only)

```
# Drop + recreate (when DataSeeder changes)
cd backend
dotnet ef database drop --project src/Persistence --startup-project src/API --force
dotnet ef database update --project src/Persistence --startup-project src/API
```

6

Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
API returns 500 on start	DB not ready yet	Wait 45–60 s for SQL Server healthcheck to pass
Login returns 401 immediately	JWT_SECRET mismatch between API instances	Ensure same JWT_SECRET in .env on all nodes
Email not sent	SMTP FromEmail ≠ Username	Set FromEmail = Username in config (Office365 enforces this)
docker compose build fails	Node/dotnet version mismatch in Dockerfile	Check Dockerfile FROM versions match project requirements
Bookings disappear on restart	DataSeeder not idempotent	Already fixed — seeder skips if data exists
ghcr.io push 403	Missing GITHUB_TOKEN write:packages permission	Set package write permission in workflow YAML